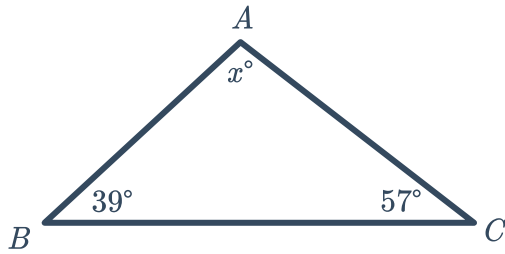


Angles of triangles

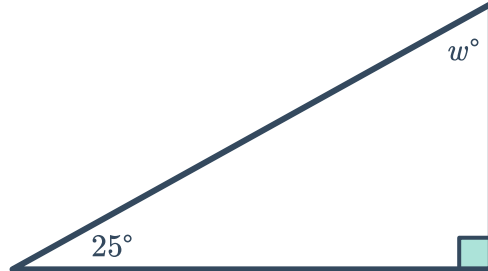


- 1 State the sum of the measure of the interior angles of any triangle.
- 2 For the following diagrams, solve for the unknown.

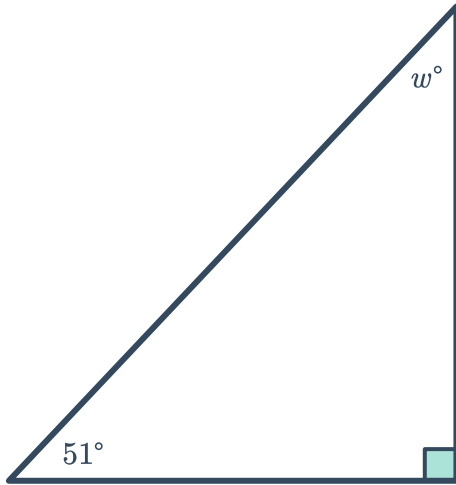
a



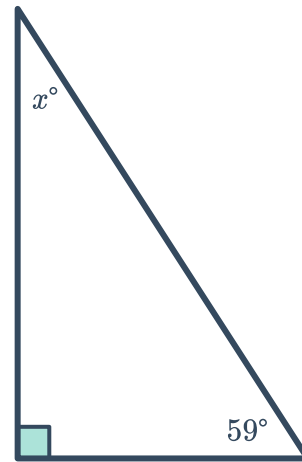
b



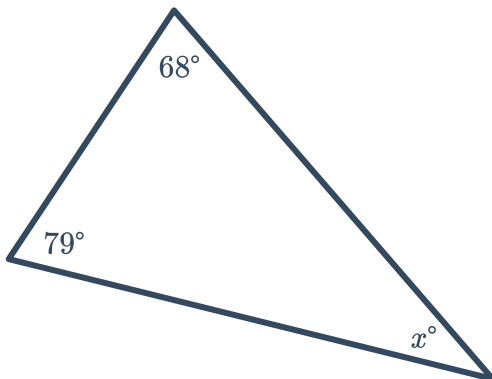
c



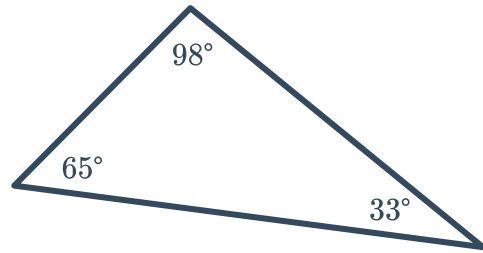
d



e



- 3 State the reason why it is not possible to make the following triangle.



- 4 Find the measure of the third angle of a triangle whose two angles measure 72° and 58° .

- 5 Classify the triangle that has two angles which measure 15° and 150° .

- 6 Consider a triangle whose three angles measure x° , y° and 48° . Determine whether the following statements are correct.

a $x + y = 48$

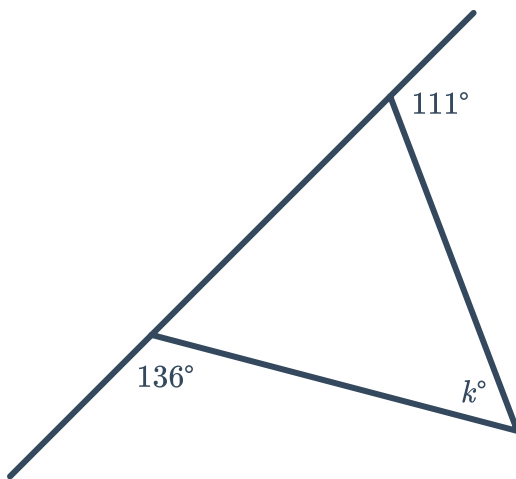
b $y = x + 48$

c $y + x = 180 - 48$

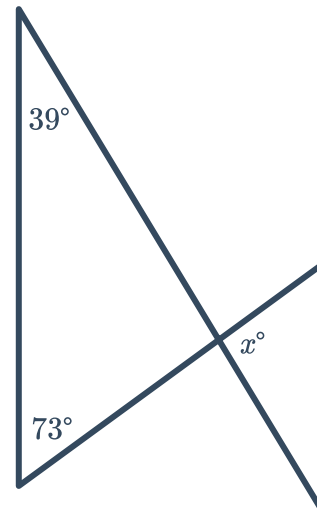
d $y = 180 - x$

- 7 For the following diagrams, solve for the value of the variable.

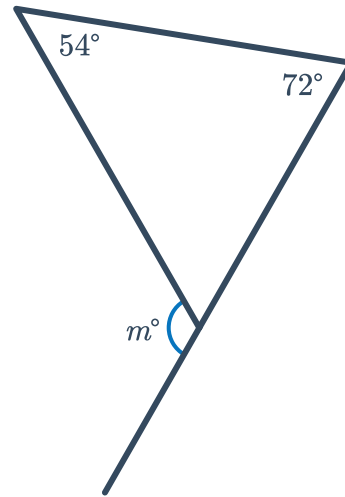
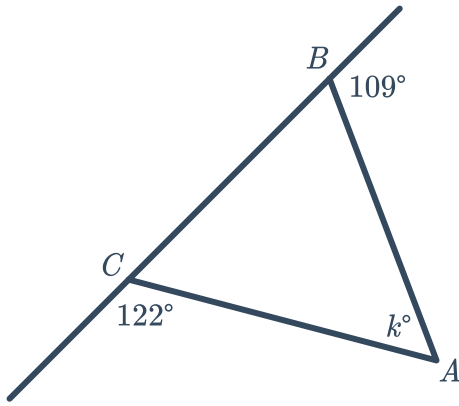
a



b

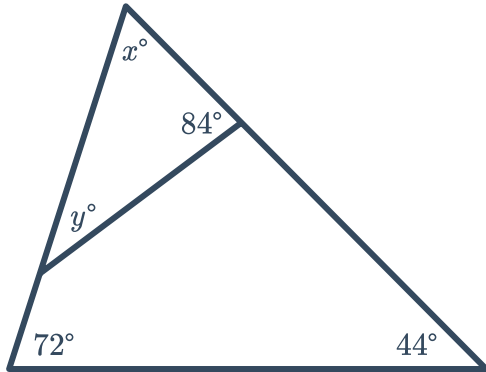


c

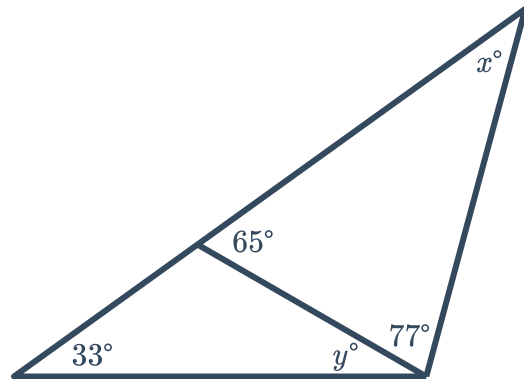


8 For the following diagrams, solve for the values of x and y .

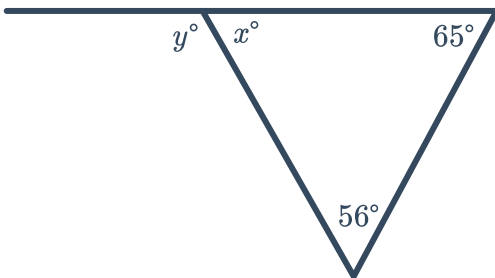
a



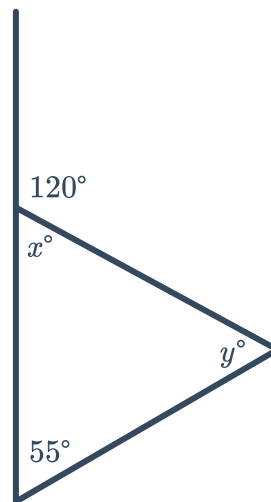
b



c



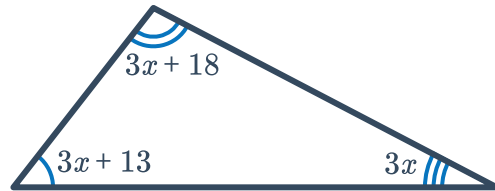
d



9 Consider the given diagram:

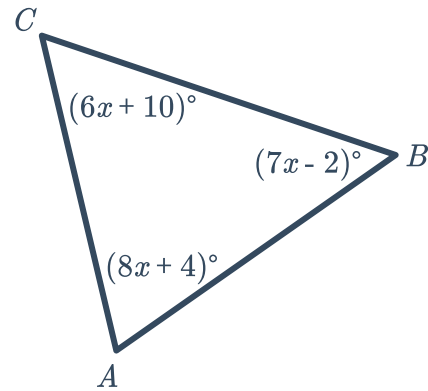


- a Solve for the exact value of x .
- b Hence, find the exact value of the smallest angle in the triangle.



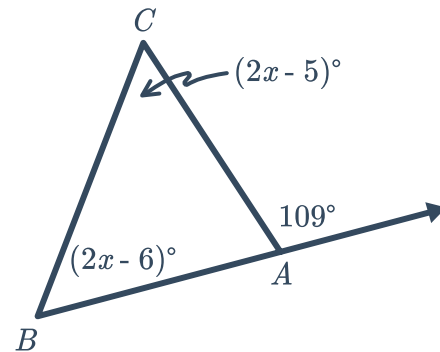
10 Consider the given diagram:

- a Solve for x .
- b Hence, find $m\angle BAC$.



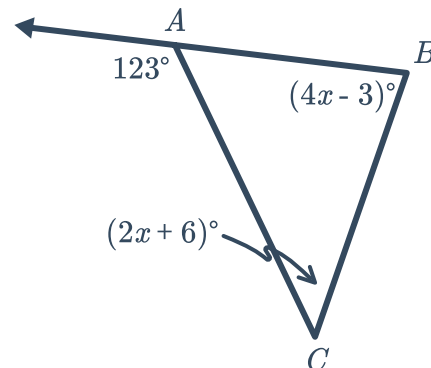
11 Consider the given diagram:

- a Solve for x .
- b Hence, find $m\angle ABC$.



12 Consider the given diagram:

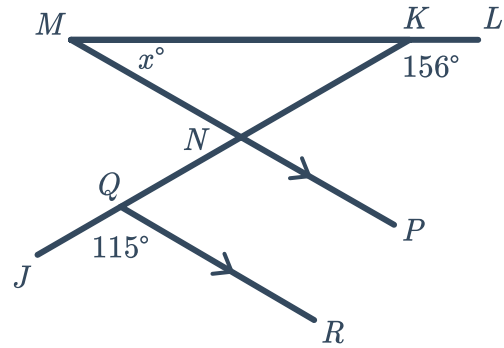
- a Solve for x .
- b Hence, find $m\angle ACB$.





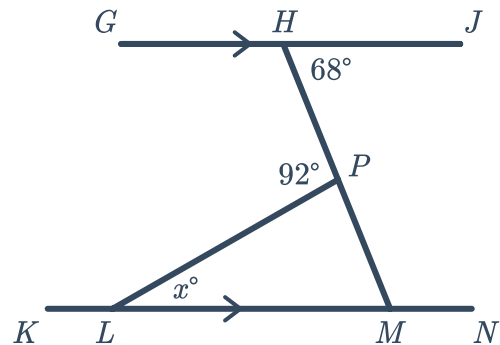
13 Consider the given diagram:

- a Describe the relationship of $\angle JQR$ and $\angle JNP$.
- b Describe the relationship of $\angle MNK$ and $\angle JNP$.
- c Hence, solve for x .



14 Consider the given diagram:

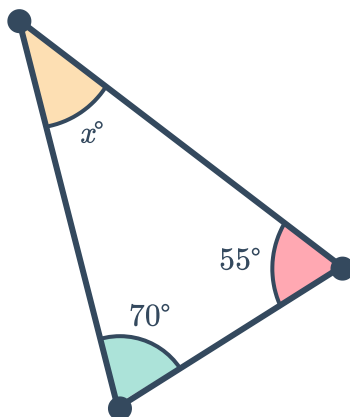
- a Describe the relationship of $\angle JHM$ and $\angle KMH$.
- b Hence, solve for x .



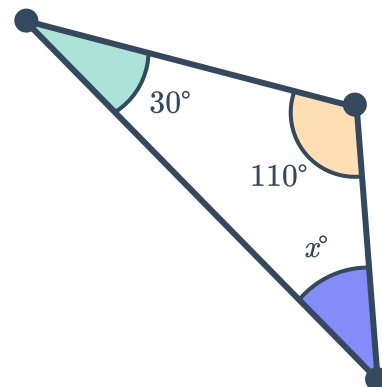
Additional questions

15 Find the value of x in the following diagrams:

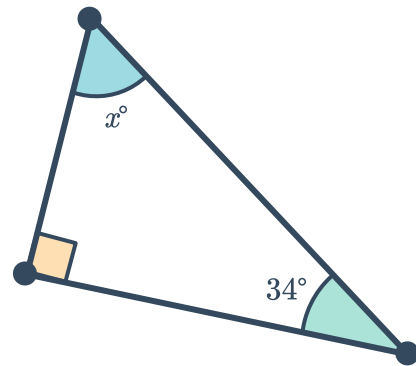
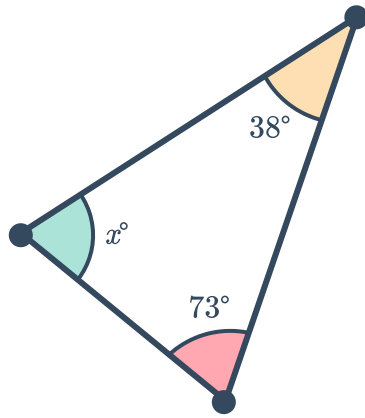
a



b



c

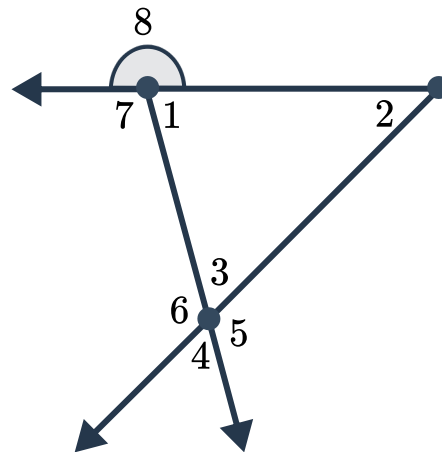


16 Find the measure of the third angle of a triangle given the measures of the other two angles:

- a** 36° and 94° **b** 45° and 45° **c** 3° and 150° **d** 75° and x°

17

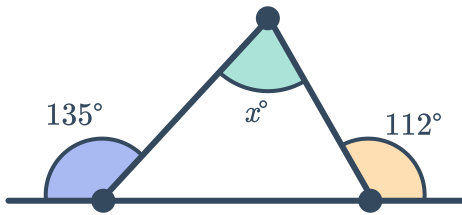
- a** Identify the numbered angles that are exterior angles.
- b** $\angle 7$ is an exterior angle for a specific interior angle. Identify the described interior angle.
- c** Describe how exterior angles 5 and 6 are related.
- d** Identify the two interior angle measures that have the same measure as angle 5? Explain your reasoning.



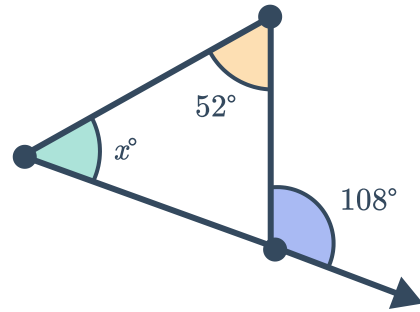
18 Find the value of x in the following diagram



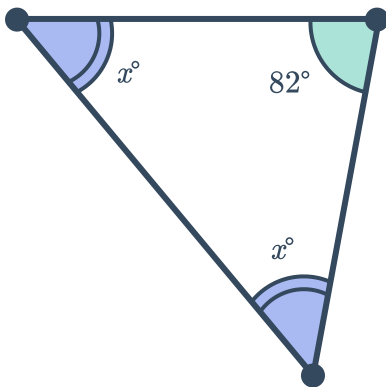
a



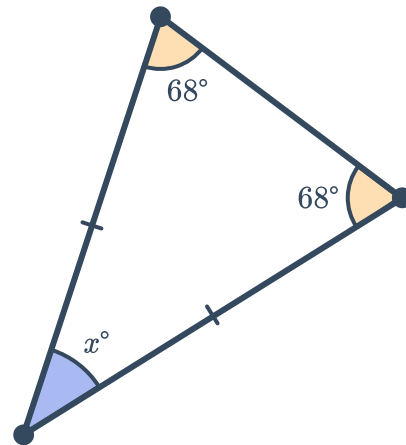
b



c



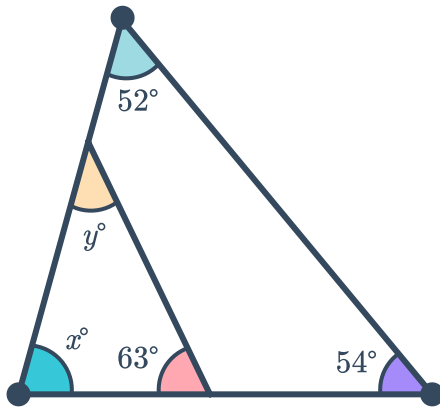
d



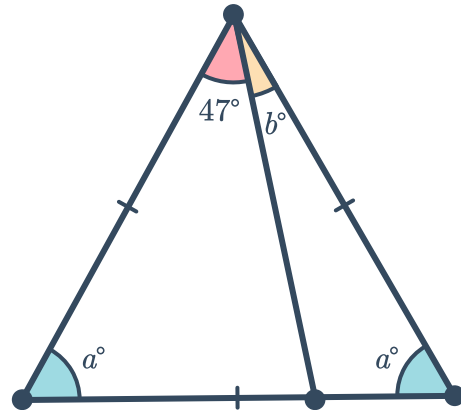
19 For the following diagrams, find the value of the variable:



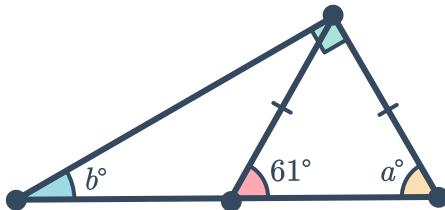
a



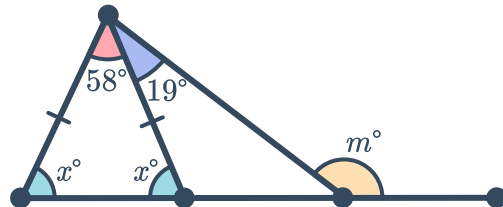
b



c



d



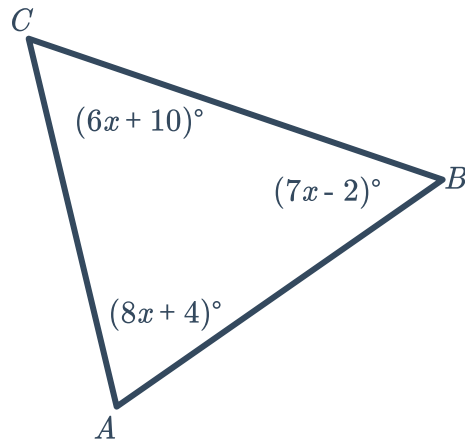
20 For each of the below diagrams:



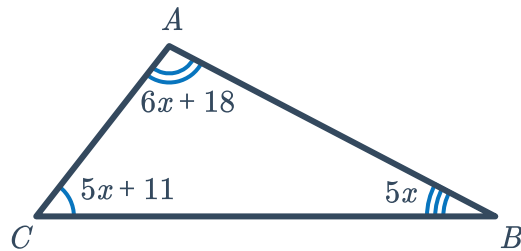
- i Find the value of x .
- ii Find the measure of $\angle CAB$.
- iii Find the measure of $\angle ABC$.
- iv Find the measure of $\angle BCA$.

Consider the given triangles and round answers to two decimal places when applicable.

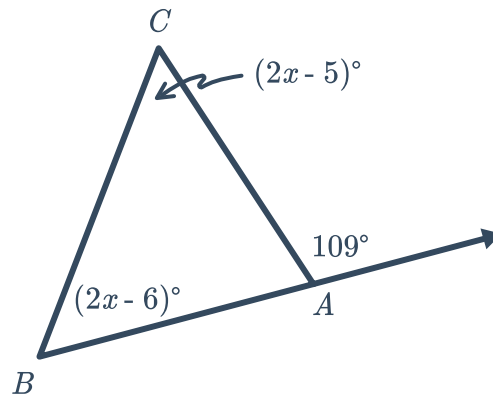
a



b



c





- 21** Determine whether the following statement is true or false. If false, provide a counterexample and correct the statement so it is true. If true, explain why.
- a** The interior angles of only right triangles sum to 180° .
 - b** The measure of an exterior angle of a triangle is equal to the sum of the two remote interior angles of the triangle.
 - c** The sum of the measures of the exterior angles of a triangle is 180° .
 - d** In a right triangle, the right angle is 90° and the remaining two angles are always congruent.
- 22** Miriam has an idea that she shares with her math class. Miriam's claim is that each pair of remote interior angles in a triangle has two exterior angles. Is this correct? Provide a diagram to support your answer.
- 23** A triangle has two angles with measures of 37 and 85. What is the measure of the smallest exterior angle of the triangle. Explain your reasoning.